

A Lagrangean-based heuristic for multi-plant, multi-item, multi-period capacitated lot-sizing problems with inter-plant transfers

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Abstract

This paper addresses scheduling of lot sizes in a multi-plant, multi-item, multi-period, capacitated environment with inter-plant transfers. A real-world problem in a company manufacturing steel rolled products provided motivation to this research. A Lagrangean-based approach, embedded with a lot shifting–splitting–merging routine, has been used for solving the multi-plant, capacitated lot-sizing problem. A “good” solution procedure developed by Sambasivan (Ph.D. Dissertation, University of Alabama, Tuscaloosa, 1994) has been used for solving the relaxed problem. About 120 randomly generated instances of the problem have been solved and it has been found that Lagrangean-based approach works quite “efficiently” for this problem.

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1. Introduction

Capacitated production lot-sizing problems pose challenges to solve owing to its combinatorial nature. Aras [1] has shown that the multi-item, capacitated lot-sizing problem with setup costs are NP-Hard. Many researchers have attempted to solve single plant, multi-item, capacitated lot-sizing problems to “very close to optimality” [2–6]. But they were not quite successful in solving large-scale problems because they could not predict the number of cutting planes that need to be generated, or the number of iterations that are required in a branch and bound approach. One group of researchers worked towards developing heuristics to solve large-scale problems [7–10]. A promising heuristic to solve large-scale problems based on Lagrangean approach was developed by Trigerio et al. [11]. This approach has gained prominence to solve capacitated lot-sizing problems. The Lagrangean relaxation

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of the capacity constraint gives rise to Wagner–Whitin problem for each item and this can be quite easily solved to optimality [12]. Column generation or sub-gradient optimization procedures can be used to update the Lagrange multipliers [13]. These problems, discussed above, deal with single plant cases.

The problem that is discussed in this research is a practical problem observed in a large steel corporation in United States of America with four manufacturing units/plants located in different parts of the country. In this research, the term “inter-plant transfer” is used as a general term. The goods can be directly sent to the customers from the plant where the goods are produced. But the customer pays only for the transportation from the nearest plant where he/she has placed the demand. The additional transportation cost from the plant where the goods have been produced has to be borne by the company. We consider this additional cost to be the same as the cost of transporting goods from the plant where the goods have been produced to the plant where the demand has been placed. Besides, a plant can get the goods from other plants and store them in inventory.

Sambasivan and Schmidt [14] have proposed a heuristic approach to solve the multi-plant, capacitated lot-sizing problem. They have developed a smoothing routine that consists of two modules to remove capacity violations: lot shifting and lot splitting. In this paper, we are investigating the efficiency of Lagrangean-based approach to solve multi-plant, capacitated lot-sizing problems. In general, Lagrangean-based approach gives solutions that are quite “close” to optimality provided the bound developed from the relaxed problem is tight and the Lagrangean-relaxed problem can be solved efficiently. Lagrangean-based approach is a popular approach to solve many practical optimization problems [15–17]. The success of the Lagrangean approach depends upon the tightness of the lower bound provided by the relaxed problem, efficiency of solving the relaxed problem, and the method of updating the Lagrangean multipliers. Lagrangean-based approach is quite easy to operationalize and it is not a surprise that it is used very widely to solve many optimization problems. This paper briefly discusses the results obtained from the uncapacitated problem and discusses in detail the Lagrangean-based heuristic for solving capacitated lot-sizing problems and the experimental setup that was used to validate the efficiency of the heuristic.

2. Mathematical model

Model MP (MP denotes multi-plant as opposed to single plant).

Sets:

- i the index denoting the product, $i = 1, \dots, n$
- j the index denoting the plant, $j = 1, \dots, m$
- t the index denoting the period, $t = 1, \dots, T$

Decision variables:

- x_{ijt} quantity of product i to be produced at plant j during period t
- I_{ijt} quantity of product i to be carried at plant j at the end of period t
- w_{ijkt} quantity of product i to be transferred from plant j to plant k during period t
- $z_{ijt} \begin{cases} 1 & \text{if there is a setup for product } i \text{ at plant } j \\ 0 & \text{otherwise} \end{cases}$

Parameters:

- d_{ijt} demand for product i at plant j during period t
 P_{jt} production capacity available at plant j during period t
 u_{ij} production rate of product i at plant j
 s_{ij} setup time rate for product i at plant j
 M_{ijt} marginal cost of producing for product i at plant j during period t
 V_{ijt} cost of setup for product i at plant j during period t
 H_{ijt} cost of carrying inventory for product i at plant j during period t
 r_{jkt} minimum cost of transferring a unit of any product from plant j to plant k during period t

Objective function:

$$\text{Minimize } \sum_i \sum_j \sum_t \left[M_{ijt}x_{ijt} + H_{ijt}I_{ijt} + V_{ijt}z_{ijt} + \left(\sum_{k \neq j} r_{jkt}w_{ijkt} \right) \right].$$

Constraints:

(1) Inventory balance constraint:

$$I_{ijt} = I_{ijt-1} + x_{ijt} - \sum_{k \neq j} w_{ijkt} + \sum_{l \neq j} w_{iljt} - d_{ijt} \quad \forall i, j, t.$$

(2) Setup forcing constraint:

$$x_{ijt} \leq \left(\sum_j \sum_{b=t}^T d_{ijb} \right) z_{ijt} \quad \forall i, j, t.$$

(3) Production capacity constraint:

$$\sum_i \left(\frac{x_{ijt}}{u_{ij}} + s_{ij}z_{ijt} \right) \leq P_{jt} \quad \forall j, t.$$

(4) Non-negativity constraints:

$$x_{ijt}, I_{ijt} \geq 0 \quad \forall i, j, t,$$

$$w_{ijkt} \geq 0 \quad \forall i, j, k \neq j, t,$$

$$z_{ijt} \in \{0, 1\} \quad \forall i, j, t.$$

The transfer cost r_{jkt} represents the minimum cost of transferring from plant j to plant k during period t . Without loss of generality, it is assumed that the transfer costs are Euclidean costs. This means that, $r_{jlt} + r_{lkt} \geq r_{jkt}$.

In the above model it can be seen that constraint (3) links all the items together. Dualizing this constraint with a set of Lagrangean multipliers, λ_{jt} , gives rise to following objective function subject to constraints (1), (2), and (4).

Minimize:

$$\sum_i \sum_j \sum_t \left[M_{ijt} x_{ijt} + H_{ijt} I_{ijt} + V_{ijt} z_{ijt} + \left(\sum_{k \neq j} r_{jkt} w_{ijkt} \right) \right] + \sum_j \sum_t \lambda_{jt} \left(\sum_i \left(\frac{x_{ijt}}{u_{ij}} + s_{ij} z_{ijt} \right) - P_{jt} \right).$$

Grouping “like” variables gives rise to:

$$\sum_i \sum_j \sum_t \left[x_{ijt} \left(M_{ijt} + \frac{\lambda_{jt}}{u_{ij}} \right) + H_{ijt} I_{ijt} + z_{ijt} (V_{ijt} + \lambda_{jt} s_{ij}) + \left(\sum_{k \neq j} r_{jkt} w_{ijkt} \right) \right] + \sum_j \sum_t \lambda_{jt} (-P_{jt}).$$

Please note that the term $\lambda_{jt} P_{jt}$ is constant for particular values of λ_{jt} and hence can be dropped from the objective function.

3. A brief discussion on multi-plant, uncapacitated lot-sizing problem

As mentioned earlier, the Lagrangean relaxation of the capacitated problem is the uncapacitated problem with modified objective function as shown above. At this juncture, it is essential to understand the theoretical foundation and solution procedure for the uncapacitated problem. However, for a detailed exposition refer to Sambasivan [18].

The key results from the uncapacitated problem are:

- (1) Multi-plant, uncapacitated lot-sizing problem can be transformed into a network problem. This result can be seen by analyzing the constraint matrix.
- (2) There exists an optimal solution such that when there is transfer to some plant j during any period t , it is from one and only one plant.
- (3) There exists an optimal solution such that if $I_{jt} > 0$ then $x_{jt} = 0$ and $w_{ljt} = 0$, $l \neq j$.
- (4) There exists an optimal solution such that if $x_{jt} > 0$ then $I_{jt} = 0$ and $w_{ljt} = 0$, $l \neq j$.
- (5) There exists an optimal solution such that if $w_{ljt} > 0$, $l \neq j$ then $I_{jt} = 0$, then $x_{jt} = 0$.

The results (3), (4), and (5) indicate the demand at a plant during any period can be satisfied either by producing at the plant, or by carrying inventory, or by transferring from an another plant.

- (6) There exists an optimal solution such that $w_{kjt} * w_{jkt} = 0$.
- (7) There exists an optimal solution such that if during period t at plant j there is a transfer into the plant, there cannot be any transfer out of plant j during period t .
- (8) Let x_{jtk_s} be the amount of demand at plant k in period s satisfied from production at plant j during period t . There exists an optimal solution where $x_{jtk_s} = 0$ or d_{ks} .
- (9) Since $x_{jt} = \sum_k \sum_s x_{jtk_s}$, production at plant j for plant k must equal $\sum_{s \in A_k} d_{ks}$, where $A_k \subseteq \{t, t+1, \dots, T\}$. This result indicates that the production quantity satisfies demand for integral number of periods.
- (10) Similarly, if $I_{jt} > 0$, then $I_{jt} = \sum_k \sum_{s \in B_{jkt}} d_{ks}$ for some $B_{jkt} \subseteq \{t+1, \dots, T\}$ and if $w_{jkt} > 0$, $k \neq j$, then $w_{jkt} = \sum_l \sum_{s \in B_{jkl}} d_{ls}$ for some $B_{jkl} \subseteq \{t+1, \dots, T\}$ and $B_{jkl} \subseteq \{t, \dots, T\}$, $l \neq j$. This result indicates that the inventory quantity satisfies demand for integral number of periods.

Based on (9) and (10), it can be seen that the transfer quantity also satisfies demand for integral number of periods.

The above properties of optimal solution do not guarantee satisfying demand for “contiguous” periods. Therefore, we make additional assumptions on the production and inventory quantities. Based on the above properties of the optimal solution and assumption that the production and inventory at any plant for any plant would satisfy demand for “contiguous” periods, the uncapacitated problem through variable redefinition has been reformulated into a special network problem, i.e., a set of shortest path problems with common fixed-charge constraints (refer to Appendix A for details on reformulation of uncapacitated problem). Each shortest path in the network corresponds to satisfying demand at a plant. A special purpose Branch and Bound approach has been used to solve the fixed-charge network problem. A logical question at this point is the need to transform into a set of shortest path problem with common fixed-charge constraints rather than as a minimum cost network. Tardos [31] has developed a strongly polynomial algorithm of the order $O(m \log n(m + n \log n))$, where n is number of nodes and m is the number of arcs, to solve minimum cost network. Our preliminary investigation revealed that if multi-plant, uncapacitated problem is transformed into a minimum cost network, the number of nodes is given by $(mT/2)(2^m + mT + m)$ and the number of arcs is given by $m(1 + 2^{m-1} + mT(1 + 2^{m-2}) + (m^2T/2)(T - 1))$, where m is the number of plants and T is the number of periods. The network becomes “messy” as the number of plants and periods increase. For example, a problem with five plants and eight periods has 1540 nodes and 5385 arcs. Therefore, the multi-plant, uncapacitated problem has been broken down into a set of shortest path problems with common fixed-charge constraints. Each shortest path network/problem corresponds to the plant whose demand is to be satisfied. The fixed charge is due to the presence of setup costs and the fixed-charge constraints are common because any plant can produce to satisfy demand at any other plant. The “break up” of a large network into small networks has made the problem easier to handle despite the introduction of fixed-charge constraints. The number of nodes in the transformed network for each plant l is given by $1 + (mT/2)(T + 3)$ and the number of arcs is given by $m(1 + T^2 + [(m - 1)T(T - 1)]/2)$. Our experimentation on algorithm for the uncapacitated problem with 216 instances reveals that our algorithm works quite efficiently despite fixed-charge constraints because the bound generated by relaxed problem (without the fixed-charge constraints) is quite tight. In all instances we could get solution within 0.25% of optimality [18].

4. Lagrangean-based heuristic for the multi-plant, capacitated lot-sizing problem

Before we discuss the Lagrangean-based procedure, let us analyze the computational complexity of the multi-plant, capacitated lot-sizing problem. After discussing the Lagrangean-based procedure, we discuss a special case of multi-plant, lot-sizing problem called parallel machine scheduling (PMS) and compare our procedure with the procedures for PMS. PMS is one of the widely researched topics in OR/MS literature.

The multi-plant capacitated problem with m plants, in terms of computational complexity, is at least as difficult as m capacitated, single plant problems. The following theorem proves this particular observation.

Table 1
Results from solving instances of capacitated problem as MILP

S. no.	Problem size ^a	Mean solution gap (%) ^b	Mean solution time (s) ^c
1	$3 \times 3 \times 5$	6.7	185
2	$3 \times 3 \times 10$	11.5	912
3	$3 \times 6 \times 5$	8.4	2349
4	$3 \times 6 \times 10$	9.4	4218
5	$4 \times 3 \times 5$	6.2	150
6	$4 \times 3 \times 10$	10.6	1332
7	$4 \times 6 \times 5$	6.6	1242
8	$4 \times 6 \times 10$	8.3	3876

^aProblem size—number of plants $\times$ number of periods $\times$ number of items.

^bSolution gap = ratio whose numerator is the difference between the best integer solution (optimal solution reached on 137 instances) and the solution to the relaxed problem and the denominator is the best integer solution.

^cSolution time = CPU time to reach optimal solution.

Theorem 1. *The multi-plant, capacitated lot-sizing problem is NP-Hard.*

The above theorem can be proved very easily, as a multi-plant problem can be decomposed into m single plant problems if the transfer variables between the plants take a value of zero. In other words, m single plant problems are a special case of multi-plant problem with m plants. It is already known that the single plant, multi-item, capacitated lot-sizing problem is NP-Hard [1,19]. This implies that multi-plant, capacitated problem is NP-Hard. An empirical study was conducted on 256 randomly generated instances of capacitated problem. These instances were solved as MILP problem using LINDO and the results of the study are given in Table 1.

The next logical question is: What is the motivation behind the integrated model? This question can be answered by examining the relationship between the multi-plant problem with m plants and the combination of m single plant problems. It is intuitive that the cost of multi-plant solution is at most equal to the cost of m single plant solutions. The following theorem proves this intuitive result.

Theorem 2. *Let $v(SP_j)$ be the optimal objective function value of the j th single plant problem and $v(MP)$ be the optimal objective value of the multi-plant problem with m plants. Then, $\sum_{j=1}^m v(SP_j) \geq v(MP)$ [14,18].*

The above theorem indicates that we may do better by looking at an integrated model. Therefore, in a multi-plant situation where transfers are allowed between plants, the optimum cost obtained by considering the lot sizes of all the plants simultaneously is at most equal to the sum of optimum costs by considering each plant individually. At this juncture, sufficient groundwork has been laid to understand the need for the heuristic for the integrated multi-plant model. This heuristic is based on the Lagrangean approach. Lagrangean-based heuristic has been quite successful for solving single-plant capacitated lot-sizing problems. Therefore, it can be argued that Lagrangean approach should work well for multi-plant case also. It should be noted that Lagrangean approach tends to work well when the sub-problem obtained after relaxing the

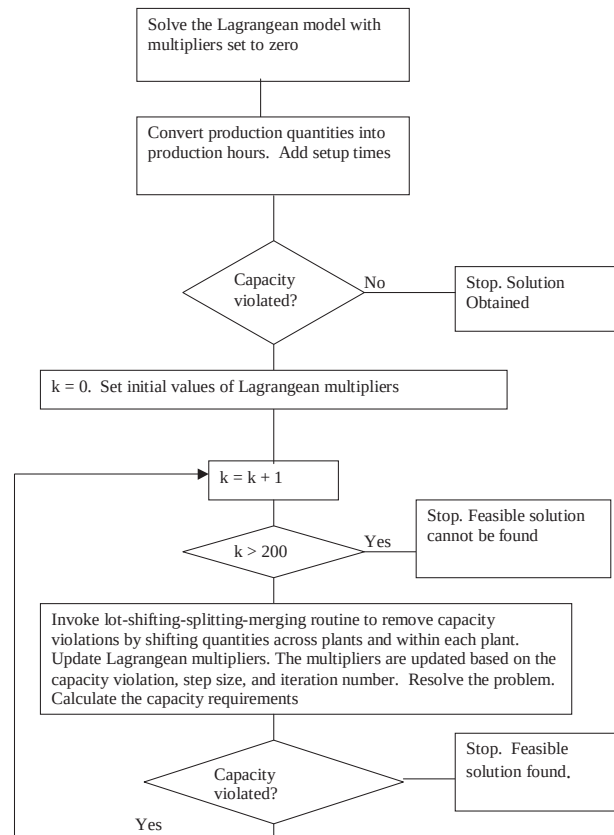


Fig. 1. Flowchart for the Lagrangean procedure.

complicating constraints (capacity) can be solved very efficiently. In the multi-plant case, it has already been shown that uncapacitated problem can be transformed into a minimum cost network and, therefore, can be solved “quite” efficiently [18]. Therefore, this research makes use of Lagrangean approach to solve the capacitated problem. The overall approach to solve the multi-plant, capacitated problem can be explained as follows and Fig. 1 illustrates the procedure in the form of a flowchart.

Step 1: Solve the uncapacitated problem for each item with the Lagrangean multipliers set to zero. The solution for each item is segregated into three parts: (1) production, (2) inventory, and (3) transfer quantities. Each of these quantities is further sub-divided according to the demand of the plant that is being satisfied. Therefore, the production, inventory, and transfer quantities can be linked to the plant whose demand is satisfied.

Step 2: The production quantities at each plant are converted to production hours by using corresponding production rates, and the production hours are added with the setup times to obtain production capacity consumption. If the production capacity consumption is greater than the production capacity available, then a capacity violation is said to exist. If there are no violations, a feasible solution is already reached and the algorithm is stopped.

Step 3: The initial value of the Lagrangean multipliers and the step size are set. Set the iteration number, $k = 0$. In this research, different initial values of Lagrangean multipliers have been tried. They are: 0, 1, 2, 3, and 4.

Step 4: A lot shifting–splitting–merging routine attempts to remove the capacity violations. This routine consists of two modules: (1) module for shifting production quantities between plants and (2) module for shifting production quantities within a plant. In the first module, the production quantities are shifted to plants where the demand is to be satisfied. The relaxed problem will have tendency to produce as much as possible in the plant that has low cost of production. Therefore, capacity violations may appear in such plants and the lot shifting procedure may remove most of the violations. If violations still persist, the second module is executed. The second module is executed for plants that have capacity violations. The production quantities are shifted within a plant across periods (forward and backward) without introducing any new violations. The production quantities are shifted to the periods where the setups have been and have not been made.

Step 5: The Lagrangean multipliers of the capacity constraints are updated using sub-gradient approach. Set $k = k + 1$. If $k > 200$, stop even if a feasible solution is not obtained and conclude that a feasible solution cannot be found.

The Lagrangean multipliers are updated by using the following expression:

$$\lambda_{jt}^{k+1} = \lambda_{jt}^k + \theta_k \frac{\sum_j \sum_t (\sum_i (x_{ijt}/u_{ij}) + s_{ij}z_{ijt}) - P_{jt}}{\|\sum_j \sum_t (\sum_i (x_{ijt}/u_{ij}) + s_{ij}z_{ijt}) - P_{jt}\|},$$

where k is the iteration number and θ is the step size. If the new λ_{jt}^{k+1} has a negative value, it is set equal to zero. In this research, two arbitrary step size rules have been used. They are $\theta_k = (\frac{1}{2})^k$ and $(\frac{3}{4})^k$. The Lagrangean relaxation is solved after updating the production and setup costs with Lagrangean multipliers as follows:

$$\text{Production cost} = M_{ijt} + \lambda_{jt}/u_{ij} \text{ and the Setup cost} = V_{ijt} + \lambda_{jt}s_{ij}.$$

Step 6: The production quantities at each plant are converted to production hours by using corresponding production rates, and the production hours are added with the setup times to obtain production capacity consumption. If violations do not exist, a feasible solution has been found and go to step 6. If not, go to step 4.

Step 7: The output for each item is compiled in terms of production quantities, inventory quantities, and transfer quantities. The total cost of production, inventory, and transfer are computed for each item. Stop.

4.1. A special case of multi-plant lot-sizing problem

In Model MP, if all r_{jkt} values are zero, then the model represents a PMS problem. Each plant corresponds to a machine. PMS is one of the widely researched areas in OR/MS literature. In the classical PMS problem, there are n jobs and m machines. Each job needs to be executed on one of the machines within a fixed processing time. So the aim is to find the schedule that optimises a certain performance measure. Most of the PMS problems are intractable and many heuristic algorithms have been developed to solve these problems [20]. The PMS problems can be generally classified into two categories: identical parallel machines and non-identical parallel machines.

Extensive research has been conducted on scheduling jobs on identical parallel machines. Ethel Mokotoff [20] has surveyed the work on identical parallel machines with the objective of minimizing makespan. She claims that scheduling problems with more than one machine involve both resource allocation and job sequencing and the complexity increases exponentially and this makes these problems intractable. Many researchers have worked on exact and heuristic approaches. Chen and Powell [21] have proposed an exact algorithm using Dantzig–Wolfe decomposition approach with the objective of minimizing an additive criterion. Ethel Mokotoff [22] has analysed polyhedral structure of the problem and has developed strong valid inequalities for fixed values of makespan. Thus, partial linear descriptions of the polytope have been used to develop an exact algorithm. LP relaxations have been solved iteratively until an integer solution is reached. Lam and Xing [23] have indicated the recent trends in PMS, such as using JIT-oriented criteria, pre-emption and setup times, and capacitated machine scheduling. Finke et al. [24] have demonstrated the use of Tabu search with simulation for solving PMS problems. Liu and Tang [32] used Genetic Algorithm to solve PMS problems with the objective of minimizing number of tardy jobs. Sevaux and Thomin [25] have computationally evaluated the heuristics and metaheuristics for PMS problems. In their research, they have discussed MILP formulation, best-insertion procedure, descent heuristics (deepest descent and multi-start descent), simulated annealing, and tabu search methods to handle PMS problems. They found that the tabu search gave them the best results.

There are numerous articles on PMS problems with non-identical parallel machines. Herbert Meyr [33] has studied PMS problem on nonidentical parallel machines with limited capacity and sequence-dependent setup times with the objective of minimizing the sum of setup, holding, and production costs. He has solved the problem heuristically by combining the local search meta-strategies and simulated annealing with dual reoptimization. Kim et al. [26] have used simulated annealing approach to schedule jobs with sequence-dependent setup times on non-identical parallel machines with the objective of minimizing total tardiness. Azizoglu and Kirca [27] studied PMS problems that involve minimizing the total cost function. They derived some properties of optimal solution and used these properties to derive a lower bound. They then devised a branch and bound algorithm to develop the schedule and the algorithm was found to solve problems up to 25 jobs in “reasonable” time. Chen and Powell [21] proposed a column generation based decomposition algorithm to solve PMS problems exactly with the objective of minimizing the total weighted earliness and tardiness. Mokotoff and Chretienne [28] devised a cutting plane algorithm for solving PMS problems with non-identical parallel machines that minimizes makespan. They identified strong valid inequalities for fixed values of maximum completion time and used these inequalities to develop an exact and approximation algorithm. Turkcan et al. [29] have solved non-identical parallel CNC machine scheduling problem with the objectives of minimizing the manufacturing cost and minimizing the total weighted tardiness. They have proposed a two-stage algorithm to find optimal machining conditions and to determine machine allocation, tool allocation, and part scheduling decisions. They combined the objectives into a single objective by using weighted linear and the Tchebycheff functions. Muller et al. [30] have used Tabu search for solving non identical PMS problems with the objective of minimizing makespan. Most of the research in PMS deals with the objective of minimizing makespan or minimizing earliness or tardiness or number of tardy jobs. In this research, the objective is to minimize the total costs. We make use of Lagrangean approach embedded with smoothing procedure to solve the multi-plant, capacitated lot-sizing problem.

5. Experimental design and computational experience

The solution procedure described above has been implemented using FORTRAN LF95. The computer implementation consists of two steps: (1) implementation of solution procedure for multi-plant, uncapacitated lot-sizing problem and (2) implementation of Lagrangean routine to obtain a feasible solution, if any.

The uncapacitated problem obtained by dropping the capacity constraint is solved for each item separately. This solution is consolidated to obtain the capacity consumption and comparison with the available capacity gives capacity violations, if any. The experimentation is divided into three steps: (1) experimental design, (2) running the instances of the problem using the algorithm described above, and (3) analysis of results.

5.1. Experimental design

The parameters that are considered in the experimental design are (a) size of the problem, (b) setup cost, (c) inventory holding cost, (d) production cost, (e) transfer cost, (f) demand, and (g) setup time.

Size of the problem: The three parameters that determine the size of the problem are number of plants, number of items, and number of periods.

The values assigned to various parameters are shown below: In Table 2, UD refers to Uniform Distribution.

5.2. Running instances of the problem using heuristic

As described earlier, each instance has been run using the algorithm for uncapacitated problem. Capacity consumptions have been computed and a Lagrangean routine has been used to remove capacity violations, if any. Different combinations of initial Lagrangean multiplier values and step size combinations have been used on 15 instances of the problem to decide on the values of Lagrangean multiplier and step size that give best results. The various combinations used were: 2 and 0.75, 2 and 0.5, 0 and 0.75, 1 and 0.5, 1 and 0.75, 4 and 0.75, and 3 and 0.75. It was found that initial

Table 2
Input parameter values

S. no.	Name of parameter	No. of values	Values
1	Plants	2	3, 4
2	Periods	4	3, 5, 6
3	Items	3	3, 5, 10, 15
4	Setup cost	1	10
5	Inventory cost	1	0–2 (UD)
5	Production cost	1	1–8 (UD)
6	Transfer Cost	1	0–2 (UD)
7	Demand (Random)	1	0–50 (UD)
8	Setup time	1	0.5

Table 3

CPU time, solution gap, CTR, and mean relaxed solution gap for various sizes of the problem

Problem size	Mean CPU time (s)	Mean solution gap (%)—Lagrangean relaxation	Mean CTR	Mean relaxed solution gap (%)—LP and Lagrangean—Lagrangean bound better by
3 × 3 × 5	7.66	1.53	1.53	5.47
3 × 3 × 10	13.46	1.32	1.32	10.68
3 × 3 × 15	11.6	2.48	1.21	12.52
3 × 4 × 5	4.4	0.94	1.44	6.50
3 × 4 × 10	31.04	2.34	1.33	9.16
3 × 4 × 15	21.22	3.07	1.23	10.00
3 × 5 × 5	3.68	4.45	1.26	9.07
3 × 5 × 10	3.42	2.51	1.36	7.49
3 × 5 × 15	15.76	2.43	1.31	9.07
3 × 6 × 5	13.62	3.39	1.37	4.61
3 × 6 × 10	11.76	2.55	1.28	6.45
3 × 6 × 15	17.68	1.48	1.35	8.02
4 × 3 × 5	8.952	4.14	1.36	1.86
4 × 3 × 10	11.74	4.96	1.37	6.00
4 × 3 × 15	10.94	1.77	1.26	5.23
4 × 4 × 5	6.3	5.62	1.48	3.00
4 × 4 × 10	27.1	2.75	1.28	6.25
4 × 4 × 15	33.54	1.99	1.22	8.50
4 × 5 × 5	24.62	3.84	1.41	4.16
4 × 5 × 10	33.14	1.75	1.35	5.25
4 × 5 × 15	20.56	3.25	1.28	5.25
4 × 6 × 5	29.68	2.34	1.74	4.66
4 × 6 × 10	30.92	1.43	1.48	6.57
4 × 6 × 15	46.1	0.70	1.26	9.30

Lagrangean multiplier value of 2 and a step size of 0.75 to give best results. The experimentation on 120 instances of the problem (five instances generated for each combination of number of plants, number periods, and number of products) has been done using these values.

The following statistics have been collected for each instance of the problem: solution to the relaxed problem, final solution value, solution time (CPU time in seconds), solution gap (closeness of the final solution to the optimal solution expressed as a percentage solution gap = (final solution – lower bound)/lower bound), number of iterations (number of iterations required by Lagrangean heuristic to find a feasible solution. Maximum number has been set at 200), and CTR (capacity tightness ratio = total capacity in units/total production quantity in units). CTR indicates how tight the capacity has been set. A value close to 1 indicates that the capacity is very tight. Please note that CTR cannot be equal to 1 since we need to take into account the setup times.

Table 3 gives the solution gap, CPU time, and CTR for different sizes of the problem. From Fig. 2, it can be seen that the CPU time increases with the size of the problem but the increase is not exponential. The maximum time is under 1 min. From Fig. 3, it can be seen that the

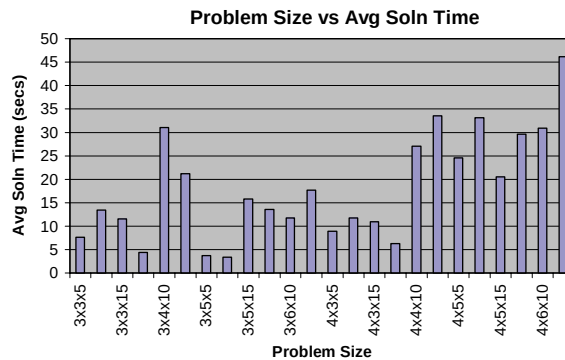


Fig. 2. Problem size vs. average solution time in s.

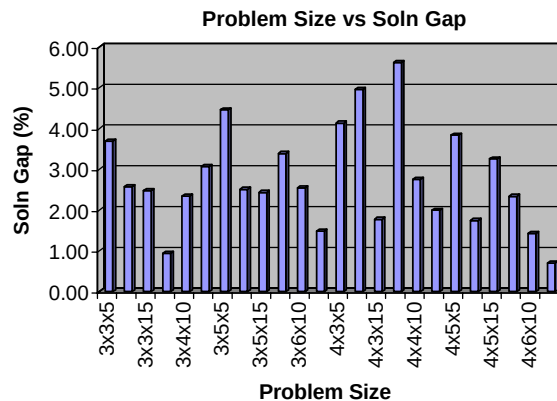


Fig. 3. Problem size vs. solution gap (%).

solution gap actually decreases with the size of the problem and the maximum solution gap is under 6%.

5.3. Analysis of results

Statistical analysis (refer to Appendix B for results) has been performed to identify the parameter(s) that causes the change in the solution gap and CPU time.

Impact of number of plants on solution gap: In this research we have used only two categories for number of plants: 3 and 4. An independent *t*-test has been performed to check if the average solution gap is same with three plants and four plants. It is observed that the average solution gap between three plants and four plants is *not* significantly different.

Impact of number of plants on CPU time: An independent *t*-test has been performed to check if the average CPU time is same with three plants and four plants. It is observed that the average CPU time between three plants and four plants is significantly different indicating that the average CPU time increases with increase in number of plants.

Impact of number of periods on solution gap: In this research, we have used four categories for number of periods: 3, 4, 5, and 6. ANOVA test has been performed to check if the average solution gap is different across number of periods. It is observed that the average solution gap is *not* significantly different across periods.

Impact of number of periods on CPU time: ANOVA test has been performed to check if the average CPU time is different across number of periods. It is observed that the average CPU time is *not* significantly different across periods.

Impact of number of items on solution gap: In this research we have three categories for number of items: 5, 10, and 15. ANOVA test has been performed to verify if the average solution gap is different across number of items. It is observed that the average solution gap across number of items is significantly different. The Post-hoc Analysis reveals that average solution gap for number of items 5 is significantly different from average solution gap for 15 (number of) items indicating that the average solution gap decreases with number of items.

Impact of number of items on CPU time: ANOVA test has been performed to verify if the average CPU time is different across number of items. It is observed that the average CPU time across number of items is *not* significantly different.

6. Conclusions

This research presented a Lagrangean-based heuristic approach to solve an integrated, multi-plant, production-planning problem that was observed in a large steel corporation in United States of America. A lot shifting–splitting–merging procedure has been embedded within the Lagrangean procedure. From the computational experience, in general, it can be concluded that Lagrangean-based heuristic approach provides good solutions for multi-plant, capacitated, lot-sizing problems with inter-plant transfers.

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Appendix A.

A.1. Details of reformulation of uncapacitated problem

The uncapacitated problem is reformulated into a network problem based on the properties of optimal solution and assumption on production and inventory quantities.

A node is uniquely represented by four parameters: i, j, k , and l with $j \leq l$. i represents the plant from where the demand for plant l is to be satisfied. j represents the current planning period. k represents a production point (a period t at any plant j is said to be a production point, if the production during the period is positive, $x_{jt} > 0$) and the first period of contiguous demand to be

satisfied at plant l . l represents the plant whose demand is to be satisfied. In the above definition, parameter k requires further explanation. The parameter k denotes the period during which the next production setup has to be made to satisfy the demand plant l . If $k = T + 1$ (T denotes number of periods in the planning horizon), then the production during period j will be sufficient to satisfy the demand through the end of planning horizon. If $j = k$ in the above node definition, then the node is called a production node. If $j < k$ in the above node definition, then the node is called a transfer/inventory node. If $i = 0$, $j = 0$, and $k = 0$, then the node is the super-source node.

A.2. Arc, cost, and other definitions

Let:

N_l denote the set of all valid nodes for plant l , $1 \leq l \leq m$, and $N_l = \{(i, j, k, l) \mid (1) i = 0, j = 0, k = 0 \text{ and } (2) 1 \leq i \leq m, 1 \leq j \leq T, j \leq k \leq T + 1\}$ and number of elements in N_l is given by $1 + (mT/2)(T + 3)$. Please note that m denotes number of plants,

- (1) Arc (p_l, q_l) denote the arc from node p_l to node q_l , where $p_l, q_l \in N_l$. Arcs will be implicitly defined by defining the immediate successors of each node. The subscript l denotes the plant whose demand is to be satisfied.
- (2) $C(i, j, l)$ denote the set of all nodes for plant i during planning period j for satisfying the demand at plant l , $C(i, j, l) = \{(i, j, j, l), (i, j, j + 1, l), \dots, (i, j, T + 1, l)\}$. Note that

$$N_l = \bigcup_{j=1}^T \bigcup_{i=1}^m C(i, j, l) \cup \{(0, 0, 0, l)\}.$$

- (3) $D(i, j, k, l)$ denote the set of immediate successors of node (i, j, k, l) . The immediate node depends upon whether the node is a super-source node or a production node or an inventory/transfer node. The possible successors to each class of node are enumerated as different cases given below:

Case 1:

$$D(0, 0, 0, l) = \{(i, 1, 1, l), i = 1, \dots, m\}.$$

Case 2:

$$D(i, j, j, l) = \{(i, j, j + 1, l), (i, j, j + 2, l), \dots, (i, j, T + 1, l)\}.$$

Case 3:

$$D(i, j, j + 1, l) = \begin{cases} \{(1, j + 1, j + 1, l), \dots, (m, j + 1, j + 1, l)\}, & j < T, \\ \{(i + 1, T, T + 1, l)\}, & j = T. \end{cases}$$

Case 4:

$$D(i, j, k, l) = \{(i', j, k, l), i' \neq i\} \cup \{(i, j + 1, k, l), k > j + 1\}.$$

(4)

$$z_{ij} = \begin{cases} 1 & \text{if there is setup at plant } i \text{ during period } j, \\ 0 & \text{otherwise.} \end{cases}$$

(5)

$$y_{p_l q_l} = \begin{cases} 1 & \text{if there is flow in the arc } (p_l, q_l), \\ 0 & \text{otherwise.} \end{cases}$$

(6) $c_{p_l q_l}$ denote the cost of traversing arc (p_l, q_l) . The node p_l is given by (i, j, k, l) and the node q_l is given by (i', j', k', l') . The cost on each arc is dependent upon the nodes the arc connects and can be calculated as given below. An explanation is also provided when the flow on the arc is 1.

Case 1: $p_l = (0, 0, 0, l)$, $q_l \in D(0, 0, 0, l)$,

$$c_{p_l q_l} = 0,$$

$y_{p_l q_l} = 1$ means that the demand at plant 1 for at least the first period will be satisfied by production at plant i' .

Case 2: $p_l = (i, j, j, l)$, $q_l \in D(i, j, j, l)$.

$$c_{p_l q_l} = \begin{cases} M_{lj} \sum_{b=j}^{k'-1} d_{lb}, & i = 1, \\ M_{lj} \sum_{b=j}^{k'-1} d_{lb} + r_{ilj} d_{lj}, & i \neq l. \end{cases}$$

When $i = 1$, $y_{p_l q_l} = 1$ means producing in plant l to satisfy its own demand in periods j through $k' - 1$. Note that where the inventory is stored is not yet committed.

When $i \neq l$, $y_{p_l q_l} = 1$ means producing in plant i to satisfy l 's demand in periods j through $k' - 1$. Note that where the inventory is stored is not yet committed.

Case 3: $p_l = (i, j, j + 1, l)$, $q_l \in D(i, j, j + 1, l)$,

$$c_{p_l q_l} = 0,$$

$y_{p_l q_l} = 1$ means that the production to satisfy the demand at plant 1 for at least period $(j + 1)$ will take place at plant i' .

Case 4: $p_l = (i, j, k, l)$, $q_l \in D(i, j, k, l)$, $j + 1 < k$.

Case 4.1: $i = i'$.

$$c_{p_l q_l} = \begin{cases} H_{lj} \sum_{b=j+1}^{k-1} d_{lb}, & i = l, \\ H_{lj} \sum_{b=j+1}^{k-1} d_{lb} + r_{ilj+1} d_{lj+1}, & i \neq l. \end{cases}$$

When $i = 1$, $y_{p_l q_l} = 1$ means that the demand at plant l for the periods $j + 1$ through $k - 1$ is held at plant l during period j .

When $i \neq l$, $y_{p_l q_l} = 1$ means that the demand of plant l for periods $j + 1$ through $k - 1$ is held at plant i during period j and the demand in period $j + 1$ at plant l is met by transfer from plant i in period $j + 1$. Note that where the remaining inventory is stored is not yet committed.

Case 4.2: $i \neq i'$.

$$c_{p_l q_l} = r_{i'l'} \sum_{b=j+1}^{k-1} d_{lb}.$$

$y_{p_l q_l} = 1$ means that the demand of plant l for periods $j + 1$ through $k - 1$ is transferred from plant i to i' during period j .

Case 5: $p_l = (i, T, T + 1, l)$, $q_l \in D(i, T, T + 1, l)$,

$$c_{p_l q_l} = 0,$$

$y_{p_l q_l} = 1$ means at the end of the planning horizon after satisfying the demand at plant l , there is no transfer to other plants to commit inventory. The cost on the arcs from the inventory/transfer nodes during the last planning period is zero.

The set of shortest path problems with common fixed-charge constraints can be mathematically represented as follows:

Reformulated Problem (RP):

$$\begin{aligned} \min \quad & \sum_i \sum_j V_{ij} z_{ij} + \sum_{(p,q) \in A} c_{pq} y_{pq} \\ \text{s.t.} \quad & \sum_{\substack{p=(0,0,0,l) \\ q \in D(0,0,0,l)}} y_{pq} = 1 \quad \forall l, \quad (\text{RP1.1}) \end{aligned}$$

$$\sum_{q \in D(p)} y_{pq} - \sum_{u \in S(p)} y_{up} = 0 \quad \forall l, p \in N_l \text{ and } S(p) \text{ is the set of immediate predecessors}$$

of node p , (RP1.2)

$$\sum_l \sum_{p \in (i,j,j,l)} \sum_{q \in D(p)} y_{pq} \leq m z_{ij} \quad \forall i, j, \quad (\text{RP1.3})$$

$$y_{pq} \in \{0, 1\}, z_{ij} \in \{0, 1\},$$

where A is the set of all arcs in the m shortest path problems.

Appendix B.*Results of statistical tests (significance level = 0.05)*

No. of plants	Average solution gap (%)	Standard error
<i>Number of plants vs. solution gap (independent t-test)</i>		
3	2.66	0.29
4	2.87	0.33
<i>t</i> -test	Df	Significant probability
−0.497	118	0.620
No. of plants	CPU time (s)	Standard error
<i>Number of plants vs. CPU time (independent t-test)</i>		
3	12.94	2.09
4	23.63	3.74
<i>t</i> -test	Df	Significant probability
−2.5	118	0.014
No. of periods	Average solution gap (%)	Standard error
<i>Number of periods vs. solution gap (ANOVA)</i>		
3	3.27	0.52
4	2.79	0.44
5	3.04	0.42
6	1.98	0.33
<i>F</i> -value	Df	Significant probability
1.671	(3,116)	0.177
No. of Periods	Average CPU time (s)	Standard error
<i>Number of periods vs. CPU time (ANOVA)</i>		
3	10.73	2.45
4	20.6	4.81
5	16.86	4.52
6	24.96	5.03

<i>F</i> -value	Df	Significant probability
1.943	(3,116)	0.127
No. of periods	Average solution gap (%)	Standard error
<i>Number of items vs. solution gap (ANOVA)</i>		
5	3.55	0.46
10	2.61	0.36
15	2.16	0.26
<i>F</i> -value	Df	Significant probability
3.727	(2,117)	0.027
No. of periods	Average CPU time (s)	Standard error
<i>Number of items vs. CPU time (ANOVA)</i>		
5	12.36	2.93
10	20.32	4.48
15	22.18	3.72
<i>F</i> -value	Df	Significant probability
1.920	(2,117)	0.151

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